AWS Lab: Zero-SSH Deployment

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Objective: Deploy a secure, publicly accessible web server on AWS without relying on traditional SSH key pairs or inbound Port 22 access. This is achieved by establishing a robust network foundation and utilizing AWS Systems Manager (SSM) Run Command for secure, agent-based remote execution.

PHASE 1

Network Foundation (VPC)

Establish the isolated virtual network and the specific block where the server will reside.

- Navigate to **VPC Dashboard** > **Your VPCs** > **Create VPC**.
- **Name:** `mycafe-vpc`
- **IPv4 CIDR block:** `10.0.0.0/16`
- Navigate to **Subnets** > **Create subnet**.
- **VPC:** Select `mycafe-vpc`
- **Name:** `mycafe-public-subnet`
- **IPv4 CIDR block:** `10.0.1.0/24`

PHASE 2

Internet Connectivity

Enable communication between the isolated VPC and the public internet.

- **Internet Gateway (IGW):** Create an IGW named `mycafe-igw` and attach it to `mycafe-vpc`.
- **Route Table:** Create a route table named `mycafe-rt` associated with `mycafe-vpc`.
- **Add Route:** Destination `0.0.0.0/0` → Target: `mycafe-igw`.
- **Subnet Association:** Edit subnet associations and explicitly link `mycafe-public-subnet` to this route table. (This action defines the subnet as "Public").

PHASE 3

Security Boundaries

Configure stateless subnet-level guards (NACL) and stateful instance-level guards (Security Groups).

▷ Network ACL (Subnet Level)

- Create NACL named `mycafe-nacl` in `mycafe-vpc`.
- **Inbound Rule:** Rule #100, Type: HTTP (80), Source: `0.0.0.0/0`, Action: ALLOW.
- **Outbound Rule:** Rule #100, Type: All Traffic, Destination: `0.0.0.0/0`, Action: ALLOW (Required for return traffic).
- **Association:** Explicitly associate `mycafe-public-subnet` with this NACL.

▷ Security Group (Instance Level)

- Create Security Group named `web-sg` in `mycafe-vpc`.

- **Inbound Rule 1:** Type: HTTP, Port: 80, Source: `0.0.0.0/0`.
- **Inbound Rule 2:** Type: SSH, Port: 22, Source: `Your-Specific-IP/32` (Best practice, though SSM will be used for execution).
- **Outbound:** Allow All Traffic (Default).

PHASE 4

Compute & IAM Preparation

Launch the compute resource with the critical IAM role required for Zero-SSH management.

- Navigate to **IAM** > **Roles** > **Create Role**.
- Trusted Entity: **AWS Service** > **EC2**.
- Attach Policy: `AmazonSSMManagedInstanceCore`.
- Name the role: `mycafe-ssm-role` and create it.
- Navigate to **EC2** > **Launch Instance**.
- **Name:** `mycafe-web`
- **AMI:** Amazon Linux 2023
- **Instance Type:** `t3.micro`
- **Network:** VPC: `mycafe-vpc`, Subnet: `mycafe-public-subnet`.
- **Auto-assign Public IPv4: Enable** (Critical for public access).
- **Security Group:** Select existing: `web-sg`.
- **Advanced Details:** Leave User Data blank for now. We will use SSM for guaranteed execution.

- Launch the instance. Once running, select it, go to **Actions** > **Security** > **Modify IAM role**, and attach `mycafe-ssm-role`. Wait 2 minutes for the SSM agent to register.

PHASE 5

Zero-SSH Deployment (AWS Systems Manager)

Why this method? Traditional SSH (Port 22) is frequently blocked by local ISPs or corporate firewalls. SSM Run Command uses an outbound, encrypted HTTPS (Port 443) connection initiated by the instance, bypassing all inbound firewall restrictions securely.

- Navigate to the **Systems Manager** service in the AWS Console.
- In the left navigation pane, select **Run Command**.
- Click **Run command**.
- **Command document:** Search for and select `AWS-RunShellScript`.
- **Targets:** Choose "Choose instances manually" and select the `mycafe-web` instance.
- **Command parameters:** Paste the following exact script into the text box:

```
# Install Apache Web Server dnf install -y httpd # Start and enable the service to run on boot systemctl start httpd systemctl enable httpd # Create the default web page echo "<h1>MyCafe.com is LIVE via SSM!</h1>" > /var/www/html/index.html
```

- Scroll to the bottom and click **Run**.
- Wait approximately 30-60 seconds. The output will display **Success** with a **Response code: 0**.

PHASE 6

Verification

Confirm the deployment was successful by accessing the public endpoint.

- Go to the **EC2 Dashboard** and copy the **Public IPv4 address** of the instance.
- Open a new browser tab and navigate to: `http://<Public-IPv4-Address>`
- Note: Ensure you type *`http://`* explicitly, as *`https://`* will fail (Port 443 is not yet configured).

Expected Result: The browser will render a clean page displaying:

MyCafe.com is LIVE via SSM!